

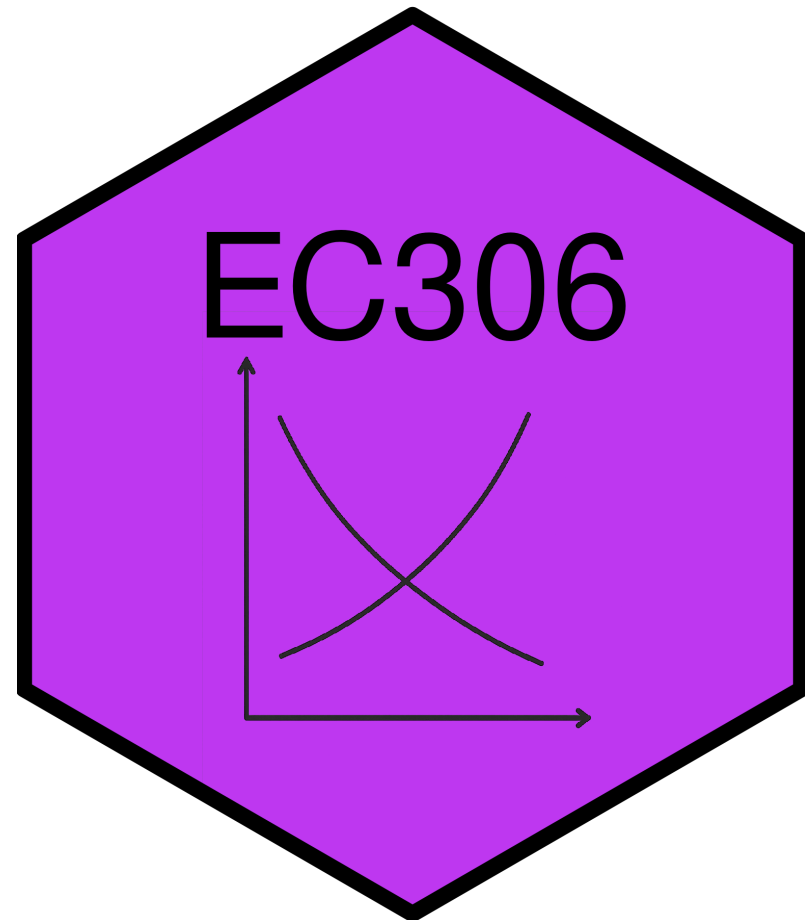
Labour Supply and Social Insurance Programs

EC306- Wages and Employment

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Goals of This Section

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In this section we will:

- Describe the main social insurance programs in Canada
- Show the effect of Employment Insurance on work decisions
- Analyze disability insurance and workers' compensation
- Examine statistical evidence for the model predictions

Unemployment Insurance

Employment Insurance: Overview

Purpose:

- Safety net for workers who lose their job
- Collect income while unemployed

Funding:

- Payroll taxes on employers and employees

Two main components:

1. Regular benefits - for those unemployed through no fault of their own
2. Special benefits - sickness, maternity, parental leave, caregiving

Employment Insurance: Benefits

Regular Benefits Structure

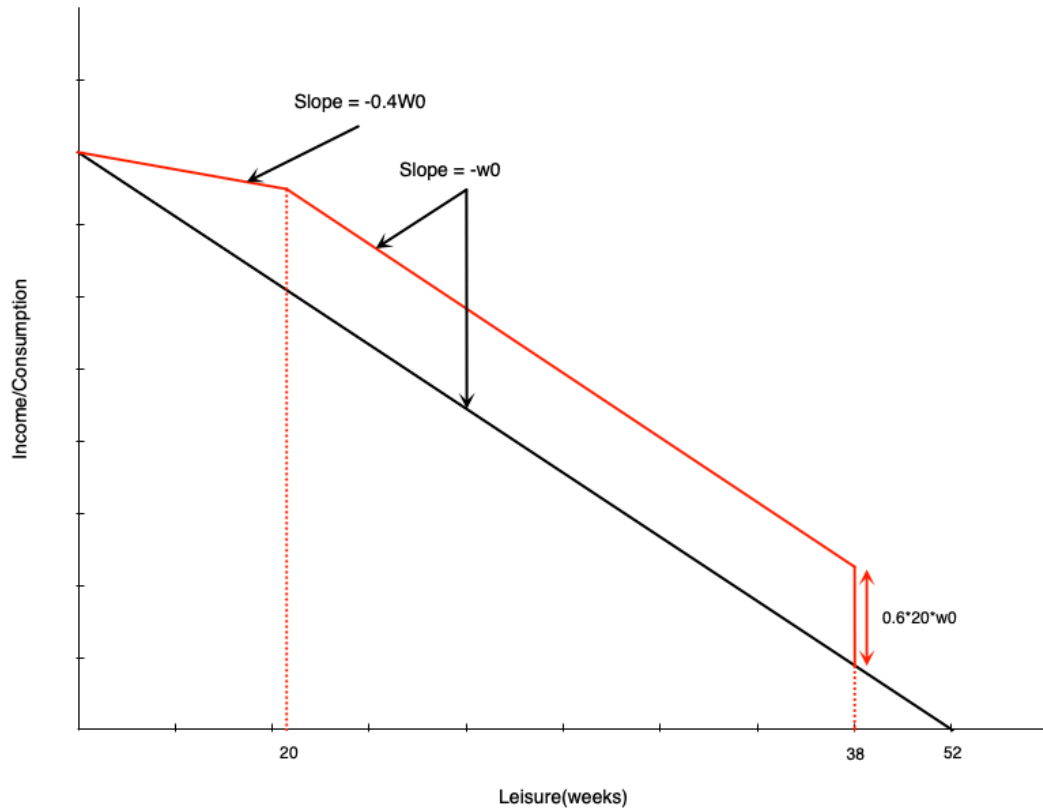
- 55% of average insurable weekly earnings
- Up to a maximum amount
- Duration: 14 to 45 weeks
 - Depends on unemployment rate and hours worked in last year

EI: Simplified Model Parameters

To analyze with income-leisure diagram:

- Must work **14 weeks** in the year to qualify
- Benefit is **60% of weekly wage** (we use 60% for simplicity)
- Maximum of **20 weeks** of benefits
- Can work up to 32 weeks and get full benefits
- Each week worked beyond 32 → lose benefits for that week
- 60% benefit = **60% tax** on earnings beyond 32 weeks
- **Cannot work and collect EI simultaneously**

El: Budget Constraint



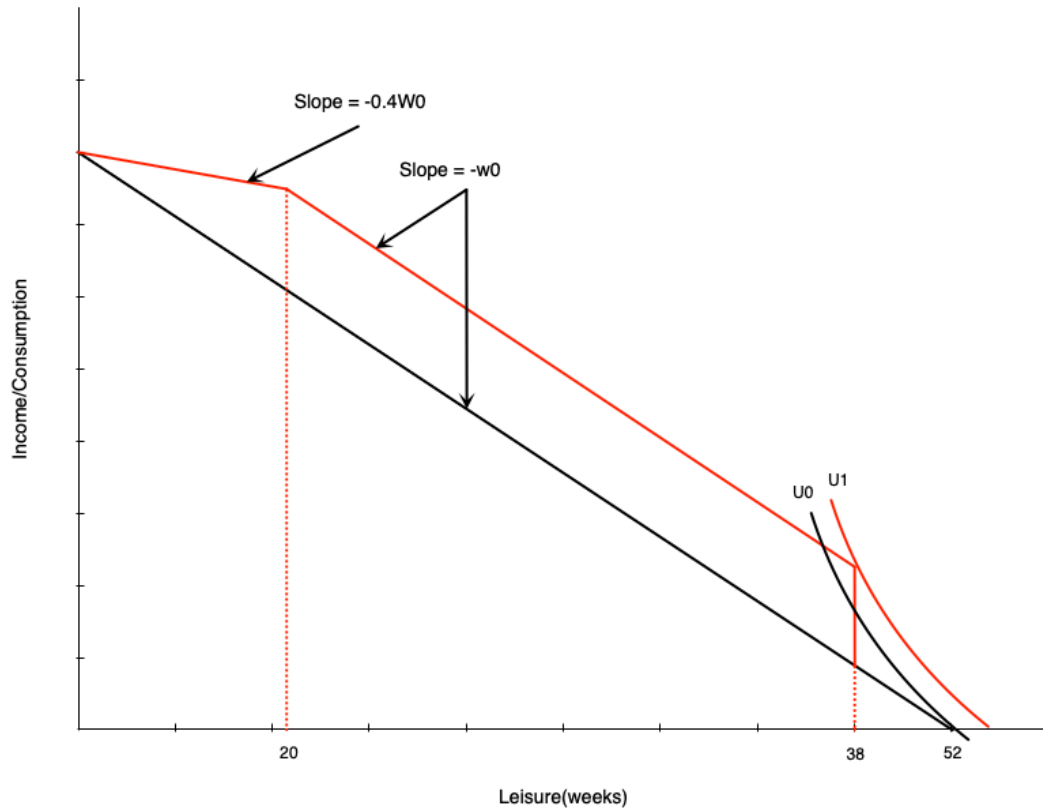
Three segments:

1. First 14 weeks: Paid at usual wage w_0
2. Weeks 14-32 of work: Get 20 weeks of benefits
 - Benefit = $0.6 \times 20 \times w_0$
3. More than 32 weeks of work: Benefits reduced week-for-week
 - Benefits paid at 60% of wage
 - Work paid at 100% of wage
 - **Keep 40% of wage** in this range

Discussion

The EI budget constraint has a segment where the effective wage is only 40% of the market wage. Why might a seasonal worker in the Maritimes choose to work exactly 14 weeks per year under this system?

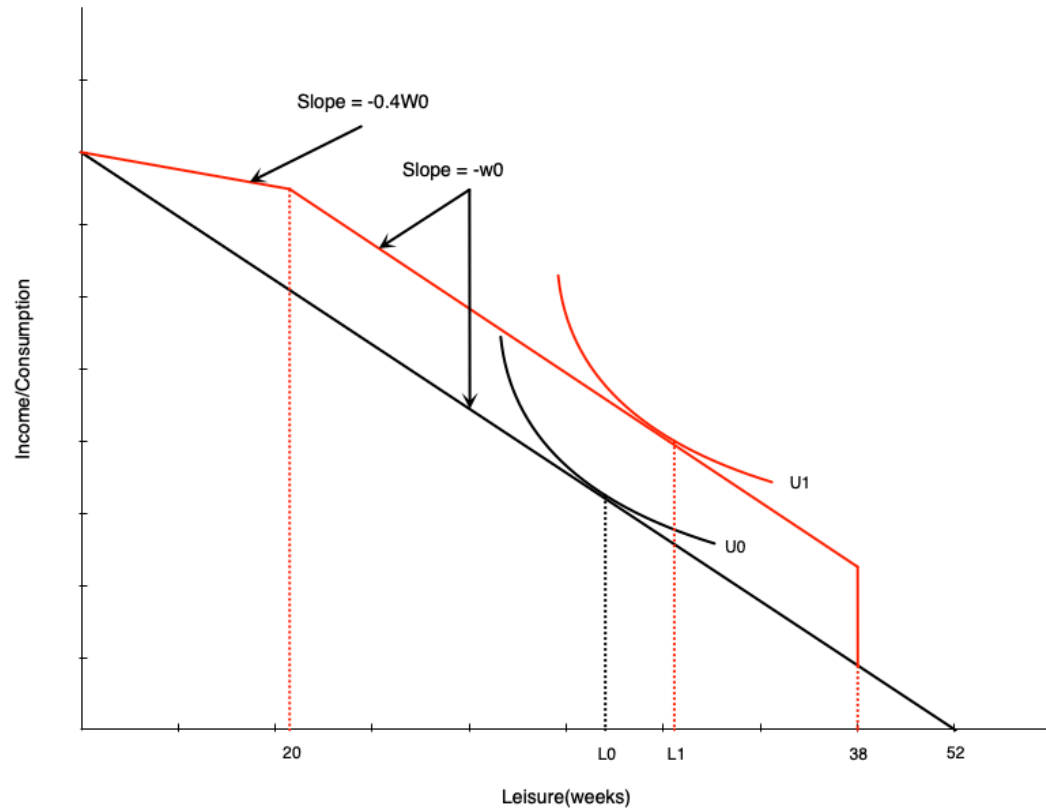
EI: Effect on Non-Participants



💡 Effect on Non-Workers

A non-participant might start working due to large increase in non-labour income after 14 weeks of work. Induces labour force entry.

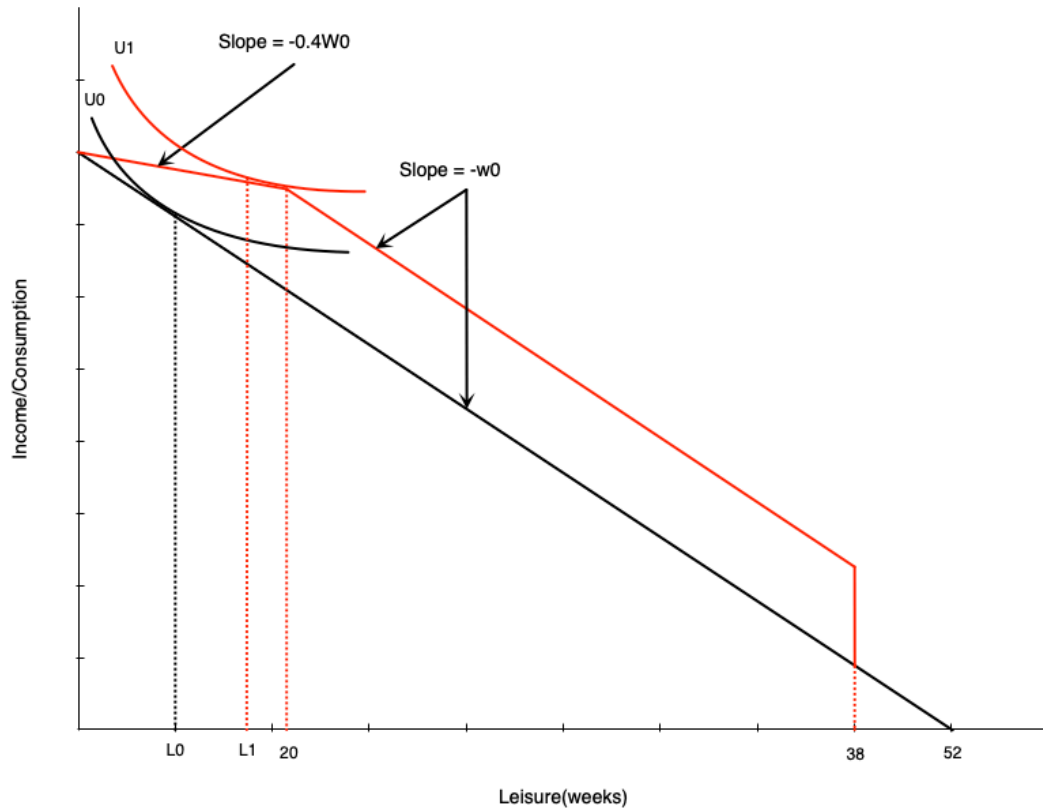
EI: Effect in Middle Range



Person getting full benefits:

- Acts like **pure income effect**
- Will reduce work weeks

EI: Effect on High-Hour Workers



In phase-out range:

- Acts like **negative income tax**
- Income effect $\rightarrow$ reduce hours $\downarrow$
- Substitution effect $\rightarrow$ reduce hours $\downarrow$

⚠ Important

Effect **unambiguously negative**

Evidence: Maine vs. New Brunswick

Natural experiment:

- Initially had very similar EI schemes
- Later, New Brunswick increased benefit generosity relative to Maine
- **Difference-in-differences approach:**
 - Compare changes in NB before/after to Maine over same period
 - Maine acts as control group
 - Controls for other time-varying factors

Evidence: Maine vs. New Brunswick Results

New Brunswick experienced:

- Higher labour force participation for marginally-attached workers
- Lower hours for those with stronger attachments

Interpretation

- Corresponds exactly to model predictions
- Highlights work disincentive effects of income maintenance programs

Disability and Workers Compensation

Introduction

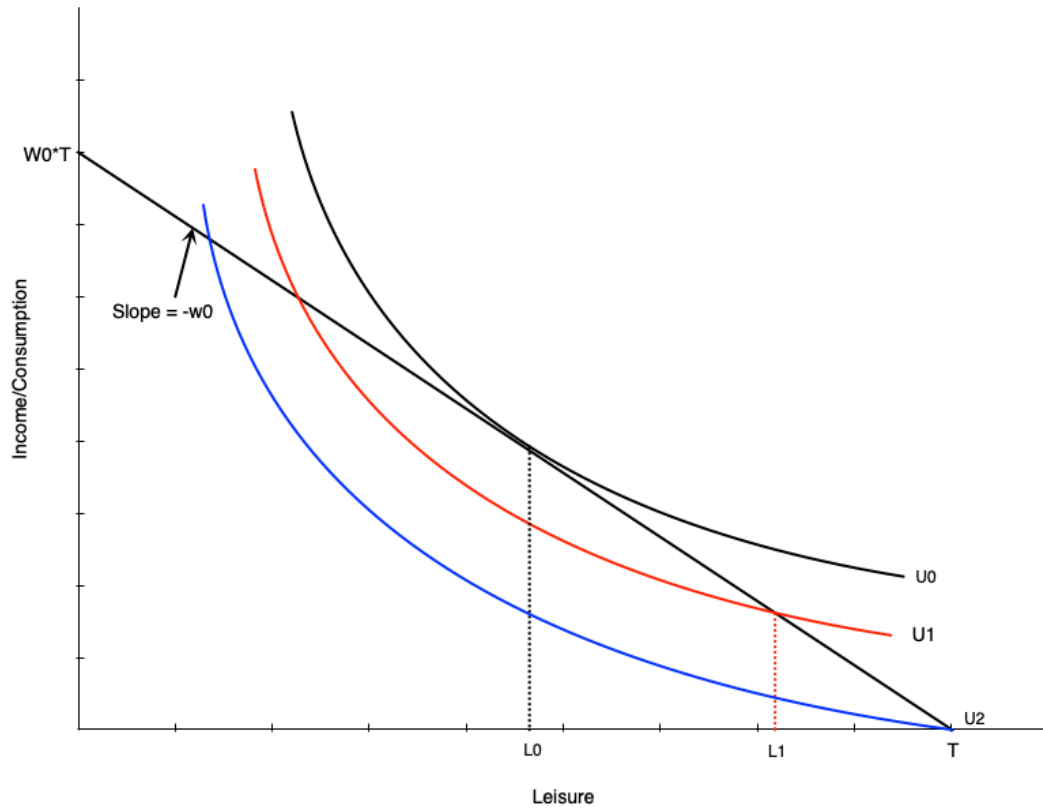
What are these programs?

- Provide income support to injured/ill workers
- Come from public and private sources
- Premiums paid by employers and employees

Work Incentive Considerations

- Severe injury/disability: work incentives not a concern (unable to work)
- Partial disability: people may still be able to work

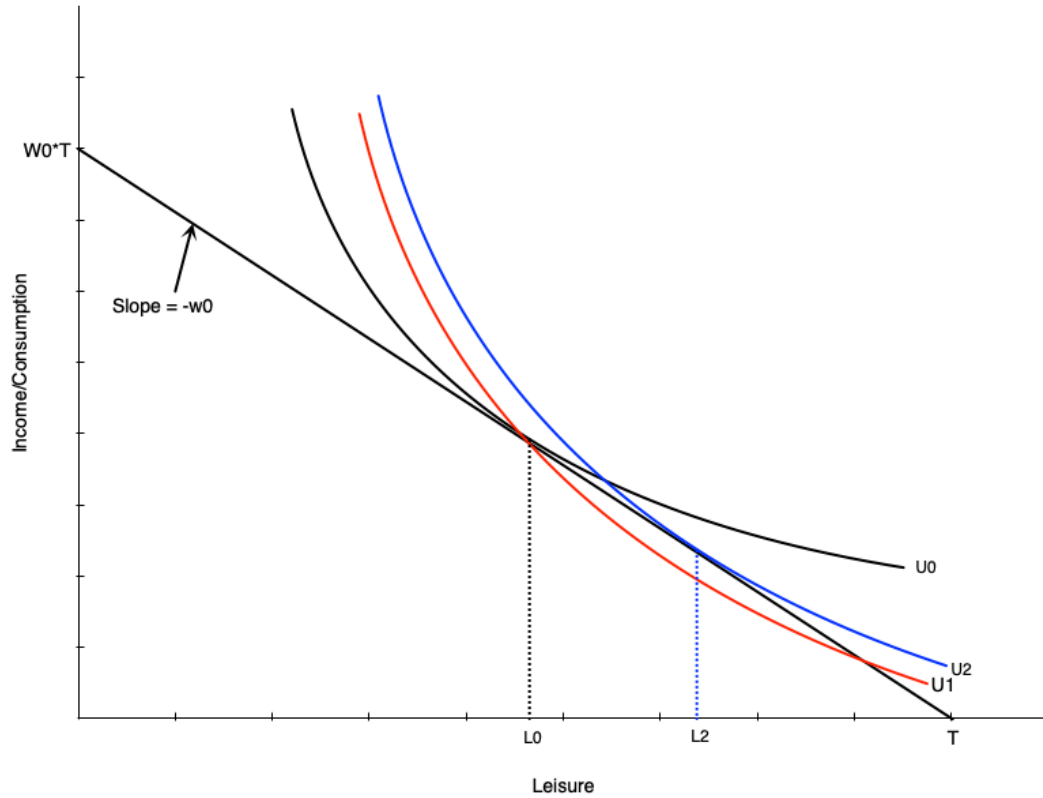
Effect of Disability on Labour Supply



Three scenarios:

1. Initially: Choose L_0 leisure, work H_0 hours
2. **Injured but can work:** Choose L_1 leisure, work H_1 hours
 - Lower (non-optimal) indifference curve
3. **Completely disabled:** Choose T leisure, zero work
 - Even lower indifference curve

Disability and Preferences



Disability may change preferences:

- Work becomes more difficult
- Preferences tilt toward leisure
- Indifference curve becomes steeper
- MRS increases at each point
- More willing to give up consumption for leisure

Result:

- Even if unconstrained, choose more leisure
- Person moves from L_0 to L_2

Compensation Schemes

Typical disability insurance structure:

- Pays a fraction of previous earnings

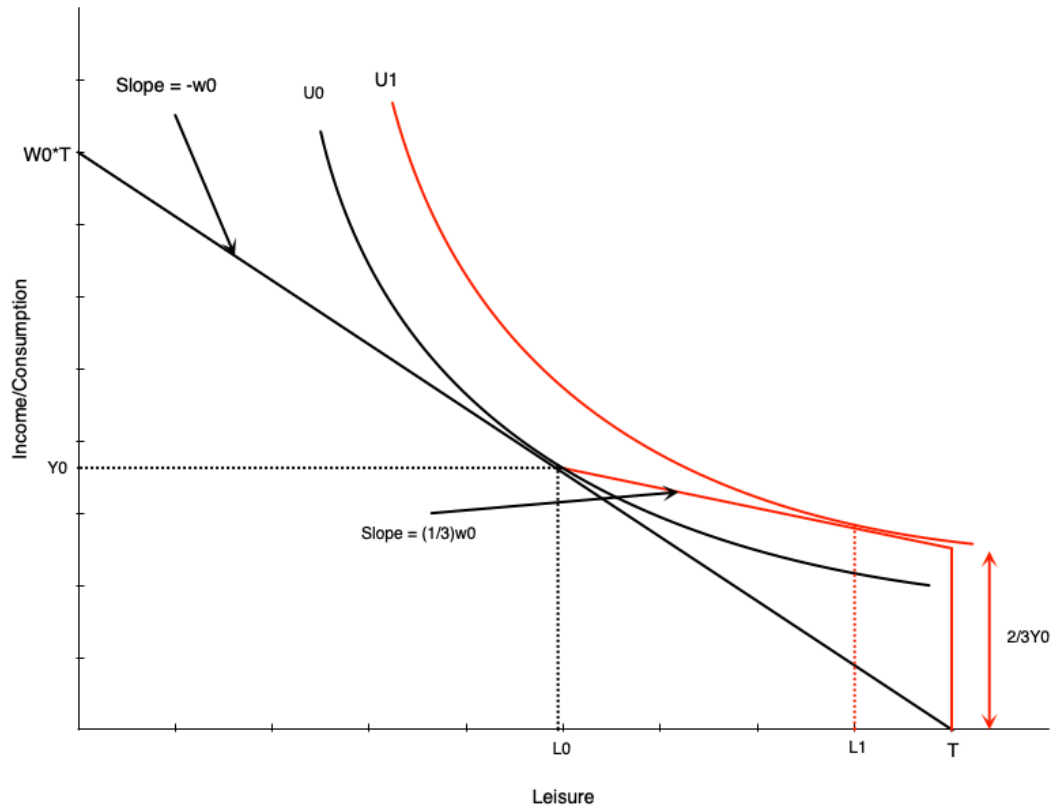
Example: Ontario WSIB

- 85% of net earnings (up to maximum)
- Lasts until return to work or age 65

Two schemes we'll examine:

1. Two-thirds income replacement
2. Full income replacement

Two-Thirds Compensation



Looks like negative income tax:

- At zero hours: get 2/3 of previous income
- For each hour worked:
 - Gain full wage
 - Lose 2/3 of wage in benefits
 - **Keep 1/3 of wage**

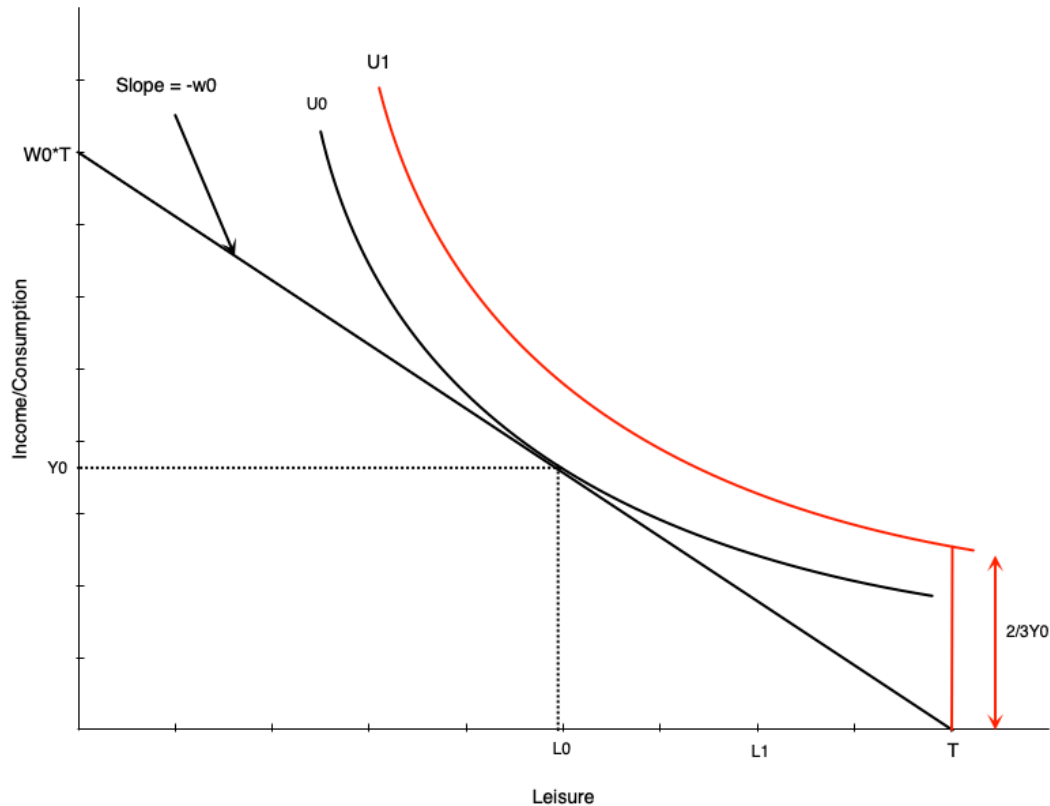
For workers with flexible hours:

- Effects match NIT
- Negative effect on hours worked

Restricted Hours: Generous Compensation

What if worker can only work H_0 hours or zero?

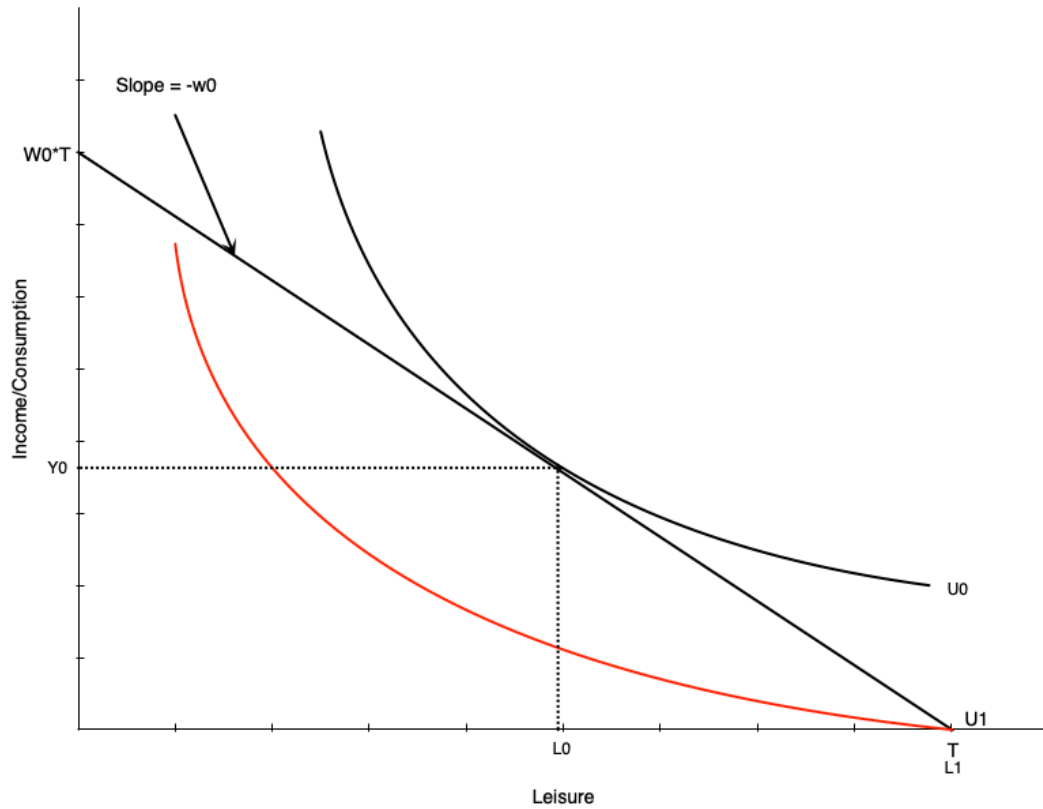
- If scheme generous enough $\rightarrow$ choose zero hours
- Utility at zero hours $>$ utility at H_0 hours



⚠ Implication

Strong work disincentives of generous scheme

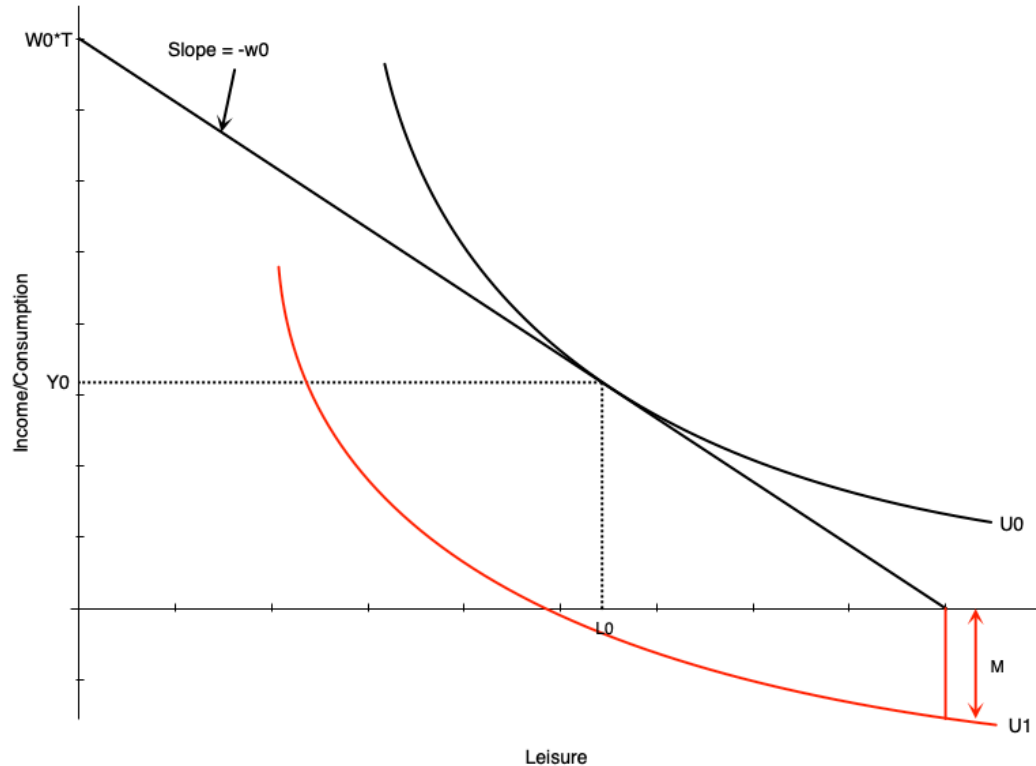
No Compensation Case



! Not Providing Compensation is Problematic

- Person who can work chooses H_0 hours
- Person who cannot work gets lower utility
- End up substantially worse off

Medical Expenses Case

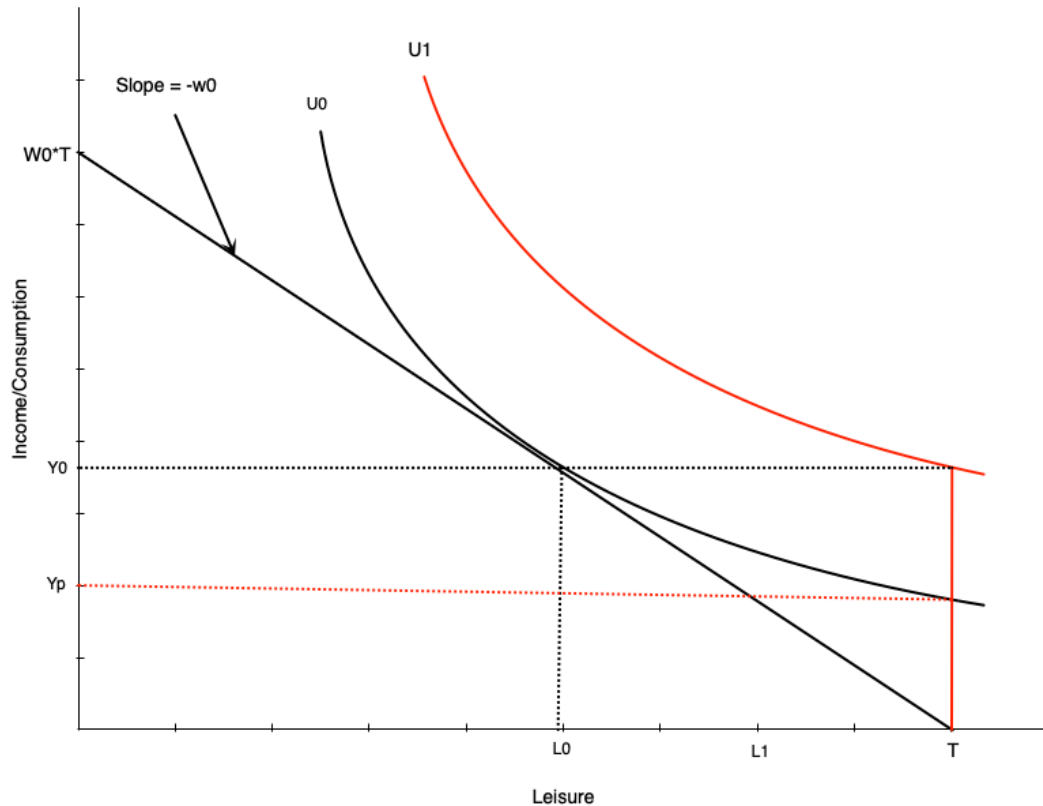


Medical expenses worsen situation:

- Act like reduction in non-labour income
- If large enough, non-labour income could go negative
- Occurs with no/very low compensation

This is an extreme scenario

Optimal Compensation Level



How much to compensate non-workers?

Two options:

1. Full income replacement

- Higher indifference curve (U_1) than working

2. Same satisfaction as working

- Same indifference curve as working
- Lower compensation (Y_p)

Note

The optimal level is a policy question

TODO — Empirical Evidence of Work Incentives (Disability)

 Not yet written

BFGLSS 2026 covers:

- Chapter 5 closes with evidence on disability insurance and workers' compensation and work incentives. The deck has the theory but no evidence slides.

Summary

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Social insurance programs:

- Insure workers against job loss, illness, and workplace injury
- Change both the budget constraint and the incentive to work

We examined:

- Employment Insurance, and its effect across the hours distribution
- Disability insurance and workers' compensation, and the trade-off in setting the compensation rate